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Comparison of umbilical-artery velocimetry and cardiotocography for surveillance of small-for-gestational-age fetuses

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Intrauterine growth retardation is associated with an increased risk of fetal asphyxia as well as greater perinatal morbidity and mortality. Ultrasound fetometry enables detection of fetuses that are small for gestational age. Doppler velocimetry of the umbilical artery has good predictive ability for fetal distress, but it is not yet clear whether it could replace cardiotocography in antenatal surveillance of small-for-gestational-age fetuses. We have done a randomised comparison of the two methods.

At four obstetric departments in Sweden, women with fetuses found to be small on ultrasound examination at 31 completed weeks of pregnancy or later were randomly assigned to antenatal surveillance with either doppler velocimetry (doppler; 214) or cardiotocography (CTG; 212). Pregnancies in the doppler group were managed according to a protocol based on blood-flow classes deriving from the semiquantitative evaluation of umbilical-artery velocity waveforms; unless the pregnancy was complicated by any other disorder, no antenatal cardiotocography was done. By comparison with the CTG group, the doppler group

had fewer monitoring occasions (mean 4.1 [SD 3.1] vs 8.2 [6.2], $p < 0.01$), antenatal hospital admissions (68 [31.3%] vs 97 [45.8%], $p < 0.01$), inductions of labour (22 [10.3%] vs 46 [21.7%], $p < 0.01$), emergency caesarean sections for fetal distress (11 [5.1] vs 30 [14.2%], $p < 0.01$), and admissions to neonatal intensive care (76 [35.5%] vs 92 [43.4%], $p = 0.10$). The groups did not differ in gestational age at birth, birthweight, Apgar scores, or total number of caesarean deliveries.

Umbilical-artery doppler velocimetry of small-for-gestational-age fetuses allows antenatal monitoring and obstetric interventions to be aimed more precisely than does cardiotocography.

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Introduction

Growth-retarded fetuses are at higher risk of asphyxia and have greater perinatal morbidity and mortality than normal fetuses.¹⁻³ It is important therefore, to identify these fetuses in utero and to monitor their wellbeing until delivery. Real-time ultrasound is widely used for intrauterine studies—eg, for fetal measurements and assessment of limb and breathing movements and amniotic fluid volume.^{4,5} The present method of choice for surveillance of small-for-gestational-age fetuses is cardiotocography (electronic fetal heart rate monitoring) used as a non-stress test.⁶ Doppler ultrasound velocimetry has lately been used for the assessment of fetal wellbeing; velocity waveform analysis of umbilical-artery blood flow can effectively detect fetal distress.⁷⁻⁹ The combination of ultrasonographic fetal measurement and umbilical-artery flow velocimetry is an effective method of fetal surveillance in high-risk pregnancy.^{10,11} In a prospective randomised study of high-risk pregnancies,¹² outcome was better when traditional methods of fetal surveillance were supplemented with umbilical-artery flow velocimetry, despite the lack of any standardised management protocol based on the results of doppler velocimetry.

Our prospective, randomised, controlled trial was designed to find out whether umbilical-artery flow velocity waveform analysis is better than cardiotocography as a method of monitoring fetal health in pregnancies complicated by intrauterine growth retardation. The study design was based on our previous experience from a series of prospective blind studies on the surveillance of small-for-gestational-age fetuses.^{9,13,14}

Subjects and methods

The four departments of obstetrics and gynaecology that took part in this 2-year study (1989–90) provide basic obstetric services for the populations of their respective geographic regions, with 3000 to 4500 deliveries each year. In all four departments, ultrasound examination is done routinely at 17–18 postmenstrual weeks for the estimation of gestational age, which is determined from the measurements of fetal biparietal diameter and femur length.¹⁵ At Karlstad and Malmö, a second ultrasound examination is done routinely at 32 weeks' gestation for the assessment of fetal growth. At Danderyd and Uppsala, a second ultrasound examination is done in the third trimester only if insufficient fetal growth is suggested by deviation in symphysis-fundal height.

The study series comprised pregnancies in which intrauterine growth retardation was suspected because a singleton fetus had an estimated weight below the mean minus 2 SD for the general population at or after 31 completed weeks of gestation. The sole exclusion criterion was fetal malformation diagnosed by ultrasound. Fetal weight was calculated from the biparietal diameter and the mean abdominal diameter measured on ultrasound examination.¹⁶ The cut-off level adopted (mean minus 2 SD) corresponds to an estimated fetal weight 22% below the expected weight for age according to the standard intrauterine growth curve for a normal Swedish population.¹⁶

Pregnant women included in the trial were randomly assigned one of two methods of antenatal fetal surveillance—umbilical-artery blood flow velocimetry (doppler group) or cardiotocography (CTG group). Randomisation was carried out by means of sealed numbered envelopes according to a randomised block design. The number of subjects needed to detect a difference was calculated with an alpha-error of 0.05 and a beta-error of 0.10, according to various primary endpoints.

Umbilical-artery blood flow velocimetry was done with a 3 MHz pulsed doppler ultrasound unit with on-line waveform analysis (GD 6, Vingmed, Horten, Norway). A 100 Hz high-pass filter was used to remove signals from slow-moving tissue in the path of the doppler ultrasound beam. Doppler signals from the umbilical cord

TABLE 1—DISTRIBUTION OF PATIENTS BY PARTICIPATING CENTRE

	Doppler group	CTG group	Total
Karlstad	64	65	129
Malmö	63	63	126
Uppsala	50	50*	100
Danderyd	37	34	71
Total	214	212	426

*Plus 1 who had to be excluded from data analysis

were recorded without imaging. The position of the sample volume was adjusted by listening to doppler signals and by visual control of the doppler spectrum displayed on the screen of the velocimeter. Only recordings during periods of fetal quiescence, without breathing or other movements, were accepted for analysis. The waveform of the maximum velocity (ie, the envelope of the recorded doppler spectrum) was analysed for pulsatility index¹⁷ and semiquantitatively evaluated for blood-flow class.¹³ 5–7 consecutive heart cycles were analysed. When the end-diastolic velocity was absent, the finding was confirmed by recording the doppler signals from at least two different angles of insonation. Doppler recordings were made by specially trained operators at each centre. The energy output of equipment used did not exceed the limits for use during pregnancy recommended by the American Institute of Ultrasound in Medicine.¹⁸ Antenatal cardiotocography was done as a non-stress test with conventional equipment (Hewlett-Packard; Sonicaid), and was assessed according to Solum et al.¹⁹

Surveillance of doppler group patients followed a strict management protocol based on the blood-flow class. Women of normal blood-flow class (positive end-diastolic velocity and pulsatility index within the gestational-age-related mean plus 2 SD of the normal population²⁰) whose fetus was of weight 22–27% below the norm were examined by doppler velocimetry every 2 weeks until delivery. A fetal weight deviation of 28–33% below the norm was taken as an indication for doppler examination every week, and a deviation of 34% or more below the norm as an indication for velocimetry twice a week and admission to hospital of the pregnant woman. Patients in blood-flow class I (positive diastolic velocity and pulsatility index $\geq$ mean plus 2 SD) were examined twice a week and those in class II (absent end-diastolic velocity) at least twice a week. A finding of blood-flow class III (absent blood flow velocity throughout diastole or reversed diastolic flow) was regarded as an indication for caesarean section on the same day. As long as there was no clinical complication other than suspected intrauterine growth retardation (ie, no pre-eclampsia or vaginal haemorrhage), patients in the doppler group did not undergo antenatal cardiotocography and they were followed up as outpatients. All patients not admitted to hospital were given leave from work, so they could reduce their physical activity. Patients in the CTG group were managed according to the clinical routines of the respective centres. Doppler velocimetry was not done for patients in this group.

Intrauterine growth was assessed every 2 weeks by ultrasound fetometry, and amniotic fluid volume was semiquantitatively assessed at the same time.²¹ During labour, patients in both groups were scheduled for monitoring by cardiotocography either intermittently or continuously. All women included in the study, which was approved by all local ethics committees, gave informed consent.

Pregnancy outcome was assessed in terms of various endpoints. Primary endpoints were gestational age at delivery, frequency of caesarean section, frequency of operative delivery for fetal distress (including emergency caesarean section and forceps and vacuum extraction), and length of stay at the neonatal intensive care unit. Secondary endpoints were number of fetal monitoring occasions, duration of antenatal hospital stay, frequency of labour induction, birthweight, frequency of small-for-dates infants, Apgar scores at 1 min and 5 min, and need for respiratory support.

Statistical analysis of the results was done with the chi-square test and Student's *t* test for unpaired observations, *p* values below 0.05 being regarded as significant. Of the 427 pregnant women who took part, 214 were allocated to the doppler group and 213 to the CTG group (table 1). No patient was lost to follow-up, but 1 CTG-group

TABLE II—CLINICAL CHARACTERISTICS OF STUDY GROUPS

—	Doppler group (n = 214)	CTG group (n = 212)
<i>Mean (SD; range)</i>		
Maternal age (yr)	27·8 (5·5; 17–44)	27·9 (5·7; 17–44)
Duration of gestation at entry (wk)	35·6 (2·2; 31–40)	35·5 (2·2; 31–41)
Estimated fetal weight deviation at entry (%)	–26·8 (4·7; –22 to –48)	–27·6 (5·8; –22 to –47)
<i>No (%) of group</i>		
Nulliparous	117 (55)	118 (56)
Smokers	98 (46)	74 (35)

patient had to be excluded from data analysis since all her records were mislaid before evaluation. 3 patients declined to take part in the trial.

Results

The two groups were similar as regards various clinical characteristics at trial entry (table II). The proportion of smokers was higher in the doppler group than in the CTG group ($p < 0·05$, chi-square test). There were no differences between the groups in maternal complications during pregnancy. The most frequent complications leading to antenatal hospital admission were pregnancy-induced or essential hypertension, pre-eclampsia, vaginal bleeding, and diabetes mellitus. The proportions of each group receiving medication (iron supplements, beta-blockers, hydralazine, tranexamic acid, or insulin) were also similar.

Fetal monitoring was done more often and more women were admitted to hospital in the CTG group than in the doppler group (table III). In the CTG group, 16 antenatal cardiotocograms were judged to indicate fetal distress; 2 improved subsequently but 14 led to delivery by emergency caesarean section. 26 cardiotocograms showed suspected fetal distress; 4 led to delivery by elective caesarean section and 9 to induction of labour. Of the 9 women whose labours were induced, 5 underwent emergency caesarean section for fetal distress.

In the doppler group, 30 (14%) patients underwent antenatal cardiotocography for clinical indications. 2 of the cardiotocograms were considered to indicate fetal distress.

TABLE III—COURSE OF PREGNANCY, LABOUR, AND DELIVERY

—	No (%) of group*		p
	Doppler group (n = 214)	CTG group (n = 212)	
<i>Mean (SD; range)</i>			
No of fetal monitoring occasions	4·1 (3·1; 0†–31)	8·2 (6·2; 1–36)	<0·001
Duration of hospital stay (per admitted patient; days)	9·0 (10·4; 1–61)	9·6 (8·4; 1–43)	0·48
<i>Patients admitted to hospital antenatally</i>	69 (31·3)	97 (45·8)	<0·01
<i>Induced labour</i>	22 (10·3)	46 (21·7)	<0·01
<i>Total operative deliveries</i>	68 (30·8)	71 (33·5)	0·56
<i>Caesarean section</i>			
Total	58 (27·1)	62 (29·2)	0·62
Elective‡	37 (17·3)	22 (10·4)	<0·05
Emergency, maternal indication	10 (4·7)	10 (4·7)	0·98
Emergency, fetal distress	11 (5·1)	30 (14·2)	<0·01
<i>Forceps and vacuum extractions</i>	8 (3·7)	9 (4·2)	0·79
<i>All operative deliveries for fetal distress</i>	30 (14·0)	50 (23·6)	<0·05

*Or mean (SD, range) as indicated
†1 patient, assigned to the doppler group, gave birth before any monitoring was done
‡Includes 15 in doppler group and 3 in CTG group for breech presentation and suspected fetopelvic disproportion

TABLE IV—CHARACTERISTICS OF INFANTS

—	Doppler group (n = 214)	CTG group (n = 212)	p
<i>Mean (SD; range)</i>			
Gestational age at birth (wk)	38·8 (2·0; 32·4–42·8)	38·6 (2·2; 31·1–43·0)	0·21
Birthweight (g)	2599 (478; 1310–3665)	2536 (538; 948–3820)	0·20
Length of stay in NICU per infant admitted (days)	12·5 (12·0; 1–63)	16·3 (15·0; 1–99)	0·12
<i>No (%) of infants</i>			
Male	92 (43·0)	92 (43·4)	0·93
Female	122 (57·0)	120 (56·6)	
Admitted to NICU	76 (35·5)	92 (43·4)	0·10
Apgar score < 7 at 1 min	8 (3·7)	16 (7·5)	0·09
Apgar score < 7 at 5 min	4 (1·9)	5 (2·4)	0·73
Requiring respiratory support	1	1	0·99

NICU = neonatal intensive care unit

In 1 case the blood-flow class was normal and elective caesarean section was done for maternal indications; the other subject had blood-flow class III and underwent emergency caesarean section according to the management protocol. No infant showed signs of asphyxia. The subgroup who underwent cardiotocography as well as doppler velocimetry did not differ significantly from the whole doppler group in terms of antenatal hospital admissions, gestation at birth, birthweight, number of induced labour, or the number of caesarean deliveries for fetal distress.

In the doppler group, 3 other patients had blood-flow class III and underwent caesarean section for this reason; they are included among the emergency caesarean sections for fetal distress (table III). 1 of the infants had a chromosomal abnormality but the other 2 are healthy.

Induction of labour owing to impaired fetal growth, pre-eclampsia, abnormal cardiotocogram, oligo-hydramnios, and psychosocial factors was significantly more frequent in the CTG than in the doppler group (table III). 144 women in the doppler group and 151 in the CTG group were monitored by cardiotocography during labour. 148 women in the doppler group and 141 in the CTG group had uneventful vaginal deliveries. The groups' overall caesarean section rates were similar, but there were fewer emergency caesarean sections for fetal distress in the doppler group than in the CTG group ($p < 0·01$).

The outcome of pregnancy was similar in the doppler and CTG groups (table IV). There were three perinatal deaths in the CTG group. 1 fetus died in utero because of placental abruption, and another was found to be dead 2 days after a normal non-stress test at 35 completed weeks of gestation; no cause of death could be found at necropsy, and the birthweight was 1900 g (27% below the norm). 1 infant with trisomy 13 died after birth. There was no perinatal death in the doppler group.

In addition to the trisomy 13 infant, 7 other infants were born with chromosomal defects. 5 had Down's syndrome (2 doppler group, 3 CTG group), 1 doppler-group baby had trisomy 18 and was treated in the neonatal ward for 21 days, and 1 CTG-group infant was born with a translocation of chromosome 18 and required neonatal care for 9 days.

Discussion

The finding of a fetus small for gestational age, as detected by ultrasound fetal measurement in the second half of gestation, gives cause to suspect intrauterine growth retardation. More intensive surveillance of such a small fetus is usual, because of the known increased risk of hypoxia in

growth-retarded fetuses. However, many of these fetuses are genetically small but otherwise healthy, and do not require longitudinal monitoring. It is highly desirable to develop a method that will enable differentiation between small but healthy fetuses and small fetuses truly at risk of intrauterine asphyxia. Doppler ultrasound velocimetry of the umbilical artery has good predictive ability for fetal distress.²² Typical changes in the blood velocity waveform, with decreased or even absent or reversed flow in diastole, are associated with poor neonatal outcome.²³

When applied to a general unselected pregnant population, doppler velocimetry did not prove practicable.²⁴ As a secondary test (eg, in a preselected group of small fetuses), it was a very efficient way of identifying fetuses at risk of distress and asphyxia.^{25,26} It has yet to be shown conclusively whether the method benefits the mother and her baby, an issue that can most reliably be settled by randomised controlled trials.²⁷

In our study, pregnancies in the doppler group were managed according to a strict protocol. The two randomly produced groups were similar for all variables examined except the frequency of smoking; maternal smoking is a well-known risk factor in pregnancy.²⁸ The differences between the groups remained and were even somewhat more pronounced when only results of non-smoking women were compared.

The study protocol deprived the doppler-group fetuses of the non-stress test, which is at present the method of choice for fetal monitoring. This study design was based on findings in previous studies that intrauterine distress was very rare among small fetuses with normal umbilical blood-flow patterns, when no complication of pregnancy other than suspected intrauterine growth retardation was present. Both open and blind prospective studies have shown that doppler velocimetry gives earlier warning of impending asphyxia than does the conventional non-stress test.^{11,29}

Since perinatal and maternal mortality rates are very low in western industrialised countries, these variables are of little use as endpoints in perinatal studies. Therefore, we chose morbidity (hospital admission of mother and infant), the frequency of obstetric intervention, and neonatal outcome. Although fetal outcome was equally good in both groups, the management protocol resulted in fewer monitoring occasions and fewer antenatal hospital admissions in the doppler group (table III). Moreover, since doppler umbilical artery velocimetry is more simple than cardiotocography, the examination takes a substantially shorter time²⁹ and thus requires less antenatal care resources.

Although the frequencies of labour induction and emergency caesarean section for fetal distress were lower in the doppler than in the CTG group, the groups did not differ in the overall incidence of caesarean section, since more women underwent elective caesarean section in the doppler group. There were significantly more operations due to breech presentation and suspected fetopelvic disproportion in the doppler than in the CTG group (15 vs 3); it is reasonable to suppose that this difference was due to chance, so we can conclude that fewer unnecessary obstetric interventions were done when clinicians were guided by umbilical circulation data than when they were guided by fetal heart rate changes.

The results were also analysed separately for the participating centres and there were similar differences between the doppler and CTG groups. As expected, the

differences did not all achieve significance owing to the small number of patients in each group. There were no significant differences between centres.

Doppler umbilical artery velocimetry and non-stress test cardiotocography examine two different aspects of fetal circulation. Our results suggest that for small-for-gestational-age fetuses in otherwise uncomplicated pregnancies doppler velocimetry can be used as the sole method of antenatal surveillance. Such management will improve the cost-effectiveness of fetal surveillance and will also diminish any maternal psychosocial stress. In other settings, however, when the risk of impaired fetal health is greater (eg, in pre-eclampsia) it is still reasonable to use the combination of both methods.

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SHORT REPORTS

5-hydroxyindoleacetic acid in cerebrospinal fluid and prediction of suicidal behaviour in schizophrenia

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Low concentrations of the serotonin metabolite 5-hydroxyindoleacetic acid (5-HIAA) in cerebrospinal fluid (CSF) are associated with suicidal behaviour in patients with depressive illness, but studies of the relation between CSF 5-HIAA and suicide in schizophrenia have been inconclusive and have not included long-term follow-up. In a prospective study, we measured 5-HIAA in CSF taken from 30 schizophrenic patients in a drug-free state, and followed these patients for 11 years. 10 patients made suicide attempts during follow-up. Suicide attempters had significantly lower concentrations of CSF 5-HIAA at initial evaluation than non-attempters (mean [SE] 6.7 [2.2] vs 23.6 [5.6] ng/ml, $p < 0.05$). Our findings provide further evidence of the relation between serotonergic dysfunction and suicide, and suggest a role for drugs with serotonergic effects in schizophrenia.

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Suicidal behaviour is common in schizophrenic patients, with up to 20% making suicide attempts at some time and up to 10% dying by suicide.¹ The determinants of this behaviour are unclear, although a history of depressed mood has been suggested.¹ Studies of patients with depressive illness have reported a link between suicidal behaviour and low concentrations of the serotonin metabolite 5-hydroxyindoleacetic acid (5-HIAA) in lumbar cerebrospinal fluid (CSF).^{2,4} This finding is supported by follow-up studies of up to 9 years after initial lumbar puncture.³

Since depression may be a factor in suicide among schizophrenic patients it is possible that suicidal behaviour is associated with a disturbance of serotonergic function in these patients. An initial report⁵ described reduced CSF 5-HIAA in 10 schizophrenics with a recent history of suicide attempt compared with 10 without such a history, but statistical significance was not reached. Further studies are contradictory, two reporting an association^{6,7} and two failing to do so.^{8,9} Two of these studies had some patients in common;^{7,8} and one relied on historical data but excluded from lumbar puncture for determination of CSF 5-HIAA patients who were actively suicidal at the time of the study,⁹ thus biasing the patient population. Proper validation of CSF 5-HIAA as a marker for suicidal behaviour in schizophrenia requires a long follow-up after initial CSF sampling, and previous studies have been restricted largely to patients with a recent history of suicidal behaviour with no substantial follow-up. We describe a prospective study.

Between 1979 and 1983, we collected lumbar CSF samples from 35 drug-free patients in Holywell Hospital, who fulfilled the Feighner criteria for schizophrenia.¹⁰ The study was approved by the local medical ethics committee. Samples were collected in the left lateral decubitus position at 0900h after bed rest and fasting for 10 h and placed in a sterile container. Patients were caffeine and nicotine free. After acidification with 6 mol/l hydrochloric acid, CSF samples were stored at -40°C before assay. The concentrations in CSF of 5-HIAA, MHPG (3-methoxy-4-hydroxyphenylethyleneglycol), and HVA (homovanillic acid), the principal central metabolites of the neurotransmitters serotonin, noradrenaline, and dopamine, respectively, were measured with high-performance liquid chromatography (HPLC). 10 patients were drug-naïve first episode cases, 3 had not received psychotropic drugs for many years, and 22 had been free of psychotropic drugs for at least 4 weeks. Their symptoms were assessed the day before lumbar puncture by the Manchester rating scale.

Within 2 years of CSF sampling, 2 patients were lost to follow-up and 1 died from natural causes. HPLC failed to give results on the CSF from 2 further patients. In December, 1991, clinical records of the remaining 30 patients (12 male, 18 female) were examined, and information on attempts at suicide and episodes of violent behaviour since collection of CSF was collated by one of us (C. K.) who had not been involved in the initial studies and had no knowledge of the CSF findings.

The patients were followed up for a mean (range) of 131 (108-146) months. No patients committed suicide but 10 made attempts (table). At the time of CSF sampling, there